Synthetic Factual Recall Experiments: Low-Frequency Linear Relations and High-Frequency Non-Affine Memorization

LMU CIS factual-recall-mechanisms

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1 Purpose

This note documents the two synthetic setups used to test the following claims:

1. **Low-frequency linear relation.** A relation supported by hidden attribute structure can become linear/affine even when each novel triple is seen only once.
2. **High-frequency non-linear/non-affine relation.** A random relation with no attribute support can be memorized when repeated often, but it does not become an affine map that generalizes to unseen subjects.

Here "linear" means that an affine map fitted from subject-side representations to object-side representations generalizes on held-out pairs. "Non-linear" is used operationally as *non-affine memorization*: the model can assign high probability to trained random triples, but affine reconstruction from the relation-conditioned representation fails and does not generalize.

2 Code and Artifacts

The experiments are implemented in:

- `experiments/factual_recall/synthetic_curriculum.py`
- CLI entry points in `experiments/factual_recall/cli.py`
- Configs in `experiments/configs/runs/`

The main result artifacts used in this writeup are:

- Low-frequency affine run: `experiments/factual_recall/results/synthetic_exp1/metrics.json`
- Low-frequency affine summary: `experiments/factual_recall/results/synthetic_exp1/summary.md`
- Matched-frequency structured/random run: `experiments/factual_recall/results/synthetic_mixed_rel`
- Matched-frequency repeat sweep: `experiments/factual_recall/results/synthetic_mixed_relations_ma`

The focused test suite passed with:

```
cd experiments
env UV_CACHE_DIR=/tmp/uv-cache MPLCONFIGDIR=/tmp/mpl-cache \
  uv run pytest tests/test_synthetic_curriculum.py -q
```

Result: `8 passed`.

3 Shared Synthetic Entity Structure

Entities are synthetic subject tokens `a_i` and object tokens `b_j`. Each subject and object is assigned a pair of hidden attributes:

$$\texttt{a_i} \mapsto (f_i, g_i), \qquad \texttt{b_i} \mapsto (f_i + \Delta_f \bmod n_f,\ g_i + \Delta_g \bmod n_g).$$

The implementation samples globally disjoint subject/object attribute pairs where possible, so the entity vocabulary has strong attribute geometry. Attribute tokens are `f_k` and `g_k`. The separator token is `x`. Training is causal language modeling: for a sequence $t_0, \ldots, t_m$, the input is $t_0, \ldots, t_{m-1}$ and the target is $t_1, \ldots, t_m$.

4 Experiment 1: Low-Frequency Structured Relation Becomes Affine

4.1 Goal

Show that a relation can be learned as a linear/affine transformation even when most relation triples are low-frequency, specifically single exposure.

4.2 Configuration

Config file:

`experiments/configs/runs/synthetic_exp1.toml`.

Quantity	Value
Seed	7
Entities	1536 subjects and 1536 objects
Attribute vocabulary	$n_f = 64,\ n_g = 64$
Attribute shift	$\Delta_f = 7,\ \Delta_g = 11$
Relations	1 structured relation `r_0`
Model	2 layers, hidden dimension 128, 4 heads
Context length	8
Embedding tying	false
Parameters	1,216,512

Table 1: Low-frequency affine experiment configuration.

4.3 Curriculum

Phase 1: attribute scaffolding plus $k = 10$ pilot triples. All subject/object attribute sentences are included repeatedly. The first $k = 10$ relation triples are also repeated.

- Attribute repeats: 8
- Pilot relation subjects: $i = 0, \ldots, 9$
- Pilot triple repeats: 256

- Total phase-1 sequences: 27,136

The concrete sentence templates are:

```
x a_i f_p g_q x          subject attribute sentence
x b_i f_r g_s x          object attribute sentence
x a_i r_0 b_i x          structured relation triple
```

Concrete examples from the saved run:

```
x a_0  f_49 g_11 x       subject a_0 attributes
x b_0  f_56 g_22 x       object b_0 attributes
x a_0  r_0  b_0  x       phase-1 pilot triple

x a_9  f_16 g_55 x
x b_9  f_23 g_2  x
x a_9  r_0  b_9  x
```

Phase 2: single-exposure novel triples. Phase 2 uses triples not present in phase 1. Each phase-2 triple is shown once:

- Unique phase-2 triples: 1024
- Phase-2 repeats: 1
- Never-trained held-out triples after phase 2: 502
- Phase-2 learning rate: 3×10^{-4}, constant during phase 2

Concrete examples:

```
x a_10 r_0 b_10 x        phase-2 single-exposure triple
x a_11 r_0 b_11 x        phase-2 single-exposure triple
x a_100 r_0 b_100 x      phase-2 single-exposure triple
```

For example, the saved attributes include:

```
a_10:  f_51 g_28         b_10:  f_58 g_39
a_11:  f_63 g_29         b_11:  f_6  g_40
a_100: f_28 g_55         b_100: f_35 g_2
```

4.4 Metrics

Two kinds of metrics are reported.

Behavioral recall. Given prompt `x`, `a_i`, `r_0`, evaluate whether the top logit is the correct object token `b_i`.

Affine representational map. Fit an affine map on seen subject/object pairs:

$$\hat{o} = Ws + b.$$

Evaluate on held-out subjects by cosine and nearest-neighbor recovery among object vectors. The run records both embedding-space and unembedding-space maps.

4.5 Results

Metric	Phase 1	End of Phase 2
Embedding affine cosine	0.2717	0.9457
Unembedding affine cosine	0.0652	0.9543
Embedding affine nearest-neighbor accuracy	–	1.0000
Unembedding affine nearest-neighbor accuracy	–	1.0000
Seen object top-1 accuracy	1.0000	0.0059
Held-out object top-1 accuracy	0.0000	0.0040

Table 2: Low-frequency structured relation results.

Finding. The expected phase shift occurs at the representational level: the relation becomes a near-perfect affine map after the single-exposure phase. Direct next-token recall does *not* emerge in this setup. Thus this run supports the claim:

$$\text{single-exposure structured triples} \Rightarrow \text{affine relational geometry},$$

but not:

$$\text{single-exposure structured triples} \Rightarrow \text{reliable top-1 factual recall}.$$

5 Experiment 2: Structured vs Random Relations at Matched Frequency

5.1 Goal

Test whether frequency alone explains affine structure by training two relations with identical exposure counts:

- R_0: structured/attribute-supported relation.
- R_1: random relation with no relationship to attributes.

The desired pattern is:

- R_0 should be more affine and more generalizable.
- R_1 should be memorized when frequent, but remain non-affine and fail on unseen subjects.

5.2 Configuration

Config file:

`experiments/configs/runs/synthetic_mixed_relations_matched.toml`.

Quantity	Value
Seed	19
Entities	192 subjects and 192 objects
Attribute vocabulary	$n_f = 32,\ n_g = 32$
Attribute shift	$\Delta_f = 3,\ \Delta_g = 5$
Relations	2: structured R_0, random R_1
Trained subject types	128
Never-seen evaluation subjects	64
Model	2 layers, hidden dimension 128, 4 heads
Context length	8
Object loss weight	4.0

Table 3: Matched-frequency structured/random configuration.

5.3 Relation Definitions

Structured relation R_0. The structured target is identity in entity index:

$$R_0(\texttt{a_i}) = \texttt{b_i}.$$

Since `b_i`'s attributes are a fixed shift of `a_i`'s attributes, this relation is supported by the hidden attribute geometry.

Random relation R_1.

$$R_1(a_i) = b_{\pi(i)}$$

where π is a random permutation generated from the run seed. This relation is not supported by the hidden attribute geometry.

Concrete examples reconstructed from the saved matched run:

```
x a_0 r_0 b_0   x        structured
x a_0 r_1 b_132 x        random

x a_1 r_0 b_1   x        structured
x a_1 r_1 b_148 x        random

x a_5 r_0 b_5   x        structured
x a_5 r_1 b_59  x        random
```

Attribute examples from the same run:

```
a_0: f_10 g_14           b_0: f_13 g_19
a_1: f_18 g_10           b_1: f_21 g_15
a_5: f_9  g_7            b_5: f_12 g_12
```

5.4 Training Data

The mixed-relation run uses:

- Attribute sentences for all entities:

```
x a_i f_p g_q x
x b_i f_r g_s x
```

- Structured relation triples:

```
x a_i r_0 b_i x
```

- Random relation triples:

```
x a_i r_1 b_{pi(i)} x
```

The main matched run repeats both relations 32 times for the same 128 trained subjects:

$$\text{structured repeats} = \text{random repeats} = 32.$$

A sweep was also run over matched repeat counts:

$$1, 2, 4, 8, 16, 32, 64.$$

5.5 Main Matched-Frequency Result at 32 Repeats

Metric	R_0 structured	R_1 random
Train top-1 accuracy	0.953	0.766
Train top-10 accuracy	1.000	1.000
Eval top-1 accuracy	0.031	0.000
Eval top-10 accuracy	0.422	0.000
Eval mean rank	27.4	181.0
Prompt affine cosine	0.322	0.048
Prompt affine nearest-neighbor accuracy	0.156	0.016

Table 4: Matched-frequency result with 32 repeats per relation.

Interpretation. At equal frequency, the structured relation is substantially more generalizable than the random relation. The random relation is partly memorized but has no held-out top-10 generalization and almost no affine structure.

Repeats	R0 train	R1 train	R0 eval@10	R1 eval@10	R0 aff. cos	R1 aff. cos	R0 aff. NN	R1 aff. NN
1	0.031	0.016	0.000	0.000	0.919	0.127	0.953	0.016
2	0.016	0.031	0.016	0.047	0.917	0.120	0.953	0.000
4	0.094	0.031	0.047	0.000	0.730	0.084	0.922	0.016
8	0.500	0.125	0.234	0.000	0.491	0.046	0.484	0.000
16	0.750	0.234	0.391	0.016	0.369	0.067	0.250	0.000
32	0.953	0.766	0.422	0.000	0.322	0.048	0.156	0.016
64	1.000	1.000	0.375	0.016	0.215	0.035	0.016	0.016

Table 5: Matched-frequency sweep over relation repetitions.

5.6 Matched Repeat Sweep

Finding. The sweep separates structure from frequency:

- At low matched frequency, R_0 is strongly affine while R_1 is not. At one repeat, R_0 affine NN accuracy is 0.953, while R_1 is 0.016.
- As frequency rises, R_1 becomes memorized. At 64 repeats, R_1 train accuracy reaches 1.000.
- Even when memorized, R_1 remains non-affine and does not generalize: at 64 repeats, R_1 eval top-10 accuracy is 0.016 and affine cosine is 0.035.
- R_0 generalizes better behaviorally than R_1, but its prompt-space affine score declines with high repetition. This suggests repeated direct training can move even a structured relation toward a more memorized prompt representation.

6 Overall Conclusions

1. **Low-frequency structured relations can become affine.** In Experiment 1, 1024 single-exposure structured triples induce a very strong affine subject-to-object map: embedding affine cosine 0.946 and affine nearest-neighbor accuracy 1.000.
2. **Affine geometry and direct recall are separable.** In Experiment 1, the affine map is strong but direct top-1 object recall remains near zero. The model learned the geometry without reliably using it as the top logit.
3. **Random relations can be memorized at high frequency without becoming affine.** In the matched-frequency sweep, random R_1 reaches perfect train accuracy at 64 repeats but retains near-zero affine/generalization metrics.
4. **Attribute support matters independently of frequency.** At the same exposure count, structured R_0 is consistently more affine and more generalizable than random R_1, especially at low repetition counts.
5. **High repetition changes the mechanism.** The cleanest affine signal for R_0 appears at low exposure. With many repeats, R_0 remains better than R_1 behaviorally, but its prompt-space affine metric declines.

7 How to Verify

From the repository root:

```
cd experiments

# Low-frequency structured affine run
env UV_CACHE_DIR=/tmp/uv-cache MPLCONFIGDIR=/tmp/mpl-cache \
  uv run factual-recall synthetic-exp1 \
  --config configs/runs/synthetic_exp1.toml \
  --device cpu --quiet

# Matched structured-vs-random run
env UV_CACHE_DIR=/tmp/uv-cache MPLCONFIGDIR=/tmp/mpl-cache \
  OMP_NUM_THREADS=4 MKL_NUM_THREADS=4 \
  uv run factual-recall synthetic-mixed-relations \
  --config configs/runs/synthetic_mixed_relations_matched.toml \
  --device cpu --quiet

# Tests
env UV_CACHE_DIR=/tmp/uv-cache MPLCONFIGDIR=/tmp/mpl-cache \
  uv run pytest tests/test_synthetic_curriculum.py -q
```

The repeat sweep was run by varying `structured_relation_repeats` and `random_relation_repeats` together over $1, 2, 4, 8, 16, 32, 64$. The saved sweep summary is:

```
experiments/factual_recall/results/
  synthetic_mixed_relations_matched/repeat_sweep_summary.md
```

8 Caveats

- The "nonlinear" claim is a geometric/mechanistic claim about non-affine memorization, not a proof that a specific nonlinear submodule alone stores the random relation.
- The strongest positive claim is about affine representational structure, not direct top-1 factual recall.
- The matched-frequency sweep shows that high repetition can reduce the measured prompt-space affine score even for the structured relation.
- Runs reported here were CPU validation runs in the shared workspace. For paper-grade claims, rerun the same configs on rho1/rho2 with multiple seeds and larger settings.